Factor Analysis

Correlation Matrix

		x1	x2	x3	x4	x5	x6
Correlation	x1	1.000	.510	.240	.423	.307	.286
	x2	.510	1.000	.296	.407	.232	.336
	x3	.240	.296	1.000	.369	.404	.271
	x4	.423	.407	.369	1.000	.347	.342
	x5	.307	.232	.404	.347	1.000	.338
	x6	.286	.336	.271	.342	.338	1.000
	x7	.394	.393	.535	.425	.489	.392
	x8	.429	.483	.378	.414	.303	.257
	x9	.433	.441	.340	.423	.222	.274
	x10	.259	.284	.258	.296	.190	.246
	x11	.503	.472	.329	.394	.339	.347
	x12	.394	.409	.279	.423	.257	.328
	x13	.396	.432	.293	.351	.314	.277
	x14	.408	.367	.346	.486	.427	.418
	x15	.516	.488	.398	.602	.337	.362
	x16	.469	.483	.382	.436	.272	.310
	x17	.398	.305	.432	.460	.546	.368
	x18	.356	.461	.352	.408	.305	.270
	x19	.388	.340	.379	.364	.261	.228
	x20	.380	.326	.294	.530	.301	.405

Correlation Matrix

		x7	x8	x9	x10	x11	x12
Correlation	x1	.394	.429	.433	.259	.503	.394
	x2	.393	.483	.441	.284	.472	.409
	x3	.535	.378	.340	.258	.329	.279
	x4	.425	.414	.423	.296	.394	.423
	x5	.489	.303	.222	.190	.339	.257
	x6	.392	.257	.274	.246	.347	.328
	x7	1.000	.451	.410	.324	.448	.429
	x8	.451	1.000	.536	.326	.587	.476
	x9	.410	.536	1.000	.357	.529	.425
	x10	.324	.326	.357	1.000	.390	.333
	x11	.448	.587	.529	.390	1.000	.530
	x12	.429	.476	.425	.333	.530	1.000
	x13	.379	.433	.371	.288	.471	.318
	x14	.478	.343	.357	.268	.447	.413
	x15	.486	.565	.519	.391	.549	.520
	x16	.412	.534	.549	.428	.598	.534
	x17	.508	.357	.353	.315	.383	.412
	x18	.414	.475	.479	.351	.542	.428
	x19	.443	.451	.429	.417	.451	.379
	x20	.448	.437	.412	.296	.452	.512

Correlation Matrix

		x13	x14	x15	x16	x17	x18
Correlation	x1	.396	.408	.516	.469	.398	.356
	x2	.432	.367	.488	.483	.305	.461
	x3	.293	.346	.398	.382	.432	.352
	x4	.351	.486	.602	.436	.460	.408
	x5	.314	.427	.337	.272	.546	.305
	x6	.277	.418	.362	.310	.368	.270
	x7	.379	.478	.486	.412	.508	.414
	x8	.433	.343	.565	.534	.357	.475
	x9	.371	.357	.519	.549	.353	.479
	x10	.288	.268	.391	.428	.315	.351
	x11	.471	.447	.549	.598	.383	.542
	x12	.318	.413	.520	.534	.412	.428
	x13	1.000	.410	.442	.401	.307	.502
	x14	.410	1.000	.518	.436	.405	.391
	x15	.442	.518	1.000	.651	.523	.520
	x16	.401	.436	.651	1.000	.454	.509
	x17	.307	.405	.523	.454	1.000	.348
	x18	.502	.391	.520	.509	.348	1.000
	x19	.457	.375	.547	.541	.462	.484
	x20	.353	.503	.560	.491	.396	.426

Correlation Matrix

		x19	x20
Correlation	x1	.388	.380
	x2	.340	.326
	x3	.379	.294
	x4	.364	.530
	x5	.261	.301
	x6	.228	.405
	x7	.443	.448
	x8	.451	.437
	x9	.429	.412
	x10	.417	.296
	x11	.451	.452
	x12	.379	.512
	x13	.457	.353
	x14	.375	.503
	x15	.547	.560
	x16	.541	.491
	x17	.462	.396
	x18	.484	.426
	x19	1.000	.460
	x20	.460	1.000

Communalities

	Initial	Extraction
x1	.430	.393
x2	.443	.406
x3	.378	.356
x4	.477	.435
x5	.427	.517
x6	.302	.279
x7	.512	.539
x8	.511	.512
x9	.451	.480
x10	.270	.254
x11	.564	.569
x12	.449	.436
x13	.397	.350
x14	.454	.441
x15	.638	.643
x16	.588	.612
x17	.508	.529
x18	.470	.461
x19	.485	.431
x20	.493	.438

Extraction Method: Maximum Likelihood.

Total Variance Explained

Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	8.779	43.895	43.895	8.251	41.255	41.255
2	1.350	6.748	50.643	.831	4.155	45.410
3	.971	4.855	55.498			
4	.888	4.440	59.938			
5	.774	3.872	63.810			
6	.742	3.708	67.518			
7	.706	3.531	71.049			
8	.655	3.274	74.324			
9	.571	2.853	77.177			
10	.543	2.717	79.894			
11	.541	2.705	82.599			
12	.506	2.530	85.129			
13	.472	2.360	87.489			
14	.436	2.180	89.669			
15	.419	2.095	91.764			
16	.377	1.887	93.651			
17	.348	1.741	95.392			
18	.332	1.662	97.054			
19	.314	1.569	98.623			
20	.275	1.377	100.000			

Extraction Method: Maximum Likelihood.

Factor Matrix[a]

	Factor 1	Factor 2
x1	.623	-.067
x2	.617	-.161
x3	.541	.251
x4	.653	.094
x5	.513	.504
x6	.489	.200
x7	.676	.287
x8	.693	-.179
x9	.660	-.212
x10	.496	-.092
x11	.733	-.179
x12	.653	-.097
x13	.589	-.057
x14	.639	.181
x15	.799	-.071
x16	.753	-.212
x17	.642	.342
x18	.665	-.136
x19	.653	-.064
x20	.662	.017

Extraction Method: Maximum Likelihood.

a. 2 factors extracted. 4 iterations required.

Goodness-of-fit Test

Chi-Square	df	Sig.
287.501	151	.000

Reproduced Correlations

		x1	x2	x3	x4	x5
Reproduced Correlation	x1	.393[a]	.395	.321	.401	.286
	x2	.395	.406[a]	.294	.388	.235
	x3	.321	.294	.356[a]	.377	.404
	x4	.401	.388	.377	.435[a]	.382
	x5	.286	.235	.404	.382	.517[a]
	x6	.291	.269	.315	.338	.351
	x7	.402	.371	.438	.468	.491
	x8	.444	.456	.330	.436	.265
	x9	.425	.441	.304	.411	.231
	x10	.315	.321	.245	.315	.208
	x11	.469	.481	.352	.462	.285
	x12	.413	.418	.329	.417	.286
	x13	.371	.372	.304	.379	.273
	x14	.386	.365	.391	.434	.419
	x15	.503	.504	.414	.515	.374
	x16	.484	.499	.355	.472	.279
	x17	.377	.341	.433	.451	.501
	x18	.424	.432	.326	.422	.273
	x19	.412	.413	.338	.421	.303
	x20	.411	.405	.362	.434	.348
Residual[b]	x1		.115	-.081	.022	.021
	x2	.115		.003	.019	-.003
	x3	-.081	.003		-.009	.000
	x4	.022	.019	-.009		-.035
	x5	.021	-.003	.000	-.035	
	x6	-.006	.066	-.043	.003	-.013
	x7	-.008	.022	.098	-.043	-.002
	x8	-.015	.027	.048	-.022	.038
	x9	.008	-.001	.036	.012	-.009
	x10	-.057	-.037	.012	-.019	-.018
	x11	.034	-.009	-.023	-.068	.054
	x12	-.020	-.009	-.051	.006	-.029
	x13	.026	.060	-.011	-.028	.041
	x14	.022	.002	-.045	.051	.008
	x15	.013	-.016	-.016	.087	-.037
	x16	-.015	-.016	.027	-.036	-.007
	x17	.021	-.036	-.001	.009	.044
	x18	-.068	.028	.026	-.014	.032
	x19	-.024	-.074	.042	-.057	-.042
	x20	-.031	-.080	-.069	.096	-.047

Reproduced Correlations

		x6	x7	x8	x9	x10
Reproduced Correlation	x1	.291	.402	.444	.425	.315
	x2	.269	.371	.456	.441	.321
	x3	.315	.438	.330	.304	.245
	x4	.338	.468	.436	.411	.315
	x5	.351	.491	.265	.231	.208
	x6	.279[a]	.388	.303	.280	.224
	x7	.388	.539[a]	.417	.385	.309
	x8	.303	.417	.512[a]	.495	.360
	x9	.280	.385	.495	.480[a]	.347
	x10	.224	.309	.360	.347	.254[a]
	x11	.322	.444	.540	.521	.380
	x12	.300	.413	.470	.451	.333
	x13	.276	.381	.418	.401	.297
	x14	.349	.484	.411	.383	.300
	x15	.376	.519	.566	.542	.403
	x16	.326	.448	.560	.542	.393
	x17	.382	.532	.383	.351	.287
	x18	.298	.411	.485	.468	.342
	x19	.307	.423	.464	.445	.330
	x20	.327	.452	.455	.433	.327
Residual[b]	x1	-.006	-.008	-.015	.008	-.057
	x2	.066	.022	.027	-.001	-.037
	x3	-.043	.098	.048	.036	.012
	x4	.003	-.043	-.022	.012	-.019
	x5	-.013	-.002	.038	-.009	-.018
	x6		.004	-.046	-.007	.021
	x7	.004		.034	.025	.015
	x8	-.046	.034		.040	-.034
	x9	-.007	.025	.040		.010
	x10	.021	.015	-.034	.010	
	x11	.024	.005	.047	.008	.010
	x12	.028	.015	.007	-.027	-6.204E-5
	x13	.001	-.002	.015	-.029	-.009
	x14	.069	-.006	-.067	-.026	-.033
	x15	-.015	-.034	-.001	-.023	-.012
	x16	-.016	-.036	-.026	.007	.035
	x17	-.014	-.023	-.026	.002	.028
	x18	-.028	.004	-.011	.011	.009
	x19	-.079	.019	-.013	-.016	.087
	x20	.078	-.004	-.019	-.021	-.030

Reproduced Correlations

		x11	x12	x13	x14	x15
Reproduced Correlation	x1	.469	.413	.371	.386	.503
	x2	.481	.418	.372	.365	.504
	x3	.352	.329	.304	.391	.414
	x4	.462	.417	.379	.434	.515
	x5	.285	.286	.273	.419	.374
	x6	.322	.300	.276	.349	.376
	x7	.444	.413	.381	.484	.519
	x8	.540	.470	.418	.411	.566
	x9	.521	.451	.401	.383	.542
	x10	.380	.333	.297	.300	.403
	x11	.569[a]	.496	.442	.436	.598
	x12	.496	.436[a]	.390	.400	.528
	x13	.442	.390	.350[a]	.366	.474
	x14	.436	.400	.366	.441[a]	.498
	x15	.598	.528	.474	.498	.643[a]
	x16	.590	.512	.456	.443	.617
	x17	.409	.386	.358	.472	.488
	x18	.512	.447	.399	.401	.541
	x19	.490	.433	.388	.406	.527
	x20	.482	.430	.388	.426	.527
Residual[b]	x1	.034	-.020	.026	.022	.013
	x2	-.009	-.009	.060	.002	-.016
	x3	-.023	-.051	-.011	-.045	-.016
	x4	-.068	.006	-.028	.051	.087
	x5	.054	-.029	.041	.008	-.037
	x6	.024	.028	.001	.069	-.015
	x7	.005	.015	-.002	-.006	-.034
	x8	.047	.007	.015	-.067	-.001
	x9	.008	-.027	-.029	-.026	-.023
	x10	.010	-6.204E-5	-.009	-.033	-.012
	x11		.034	.030	.012	-.049
	x12	.034		-.072	.013	-.008
	x13	.030	-.072		.044	-.032
	x14	.012	.013	.044		.020
	x15	-.049	-.008	-.032	.020	
	x16	.008	.022	-.054	-.007	.034
	x17	-.026	.026	-.051	-.067	.035
	x18	.030	-.020	.103	-.010	-.021
	x19	-.039	-.054	.068	-.031	.020
	x20	-.030	.082	-.036	.077	.033

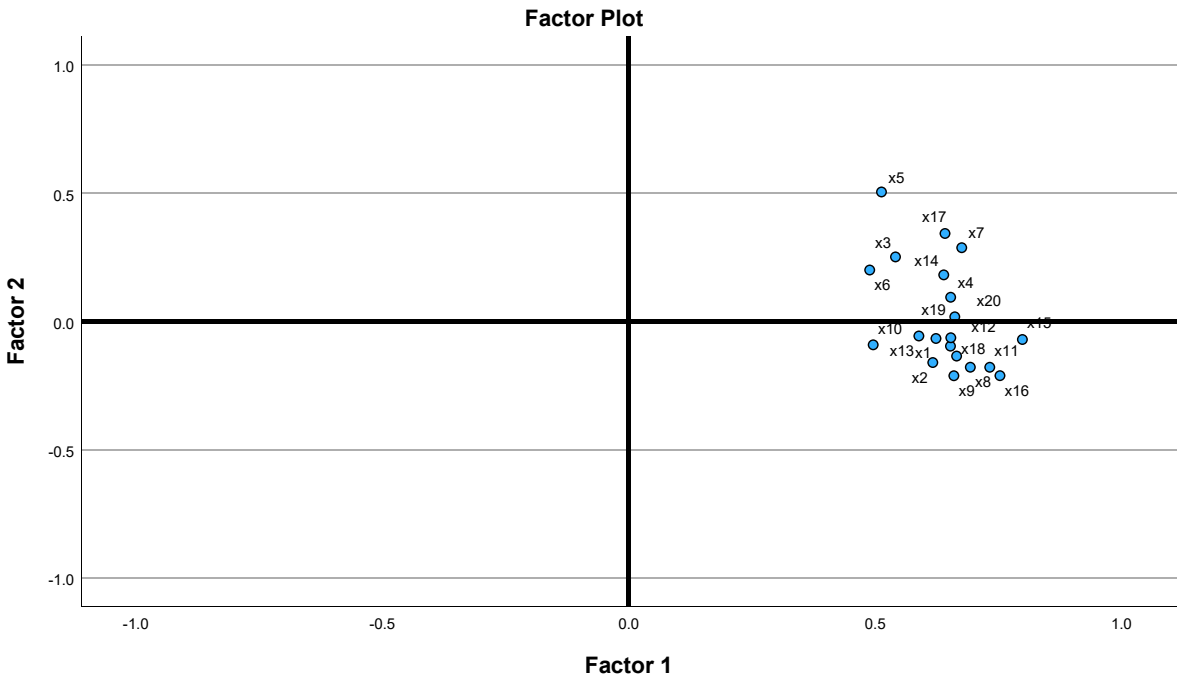
Reproduced Correlations

		x16	x17	x18	x19	x20
Reproduced Correlation	x1	.484	.377	.424	.412	.411
	x2	.499	.341	.432	.413	.405
	x3	.355	.433	.326	.338	.362
	x4	.472	.451	.422	.421	.434
	x5	.279	.501	.273	.303	.348
	x6	.326	.382	.298	.307	.327
	x7	.448	.532	.411	.423	.452
	x8	.560	.383	.485	.464	.455
	x9	.542	.351	.468	.445	.433
	x10	.393	.287	.342	.330	.327
	x11	.590	.409	.512	.490	.482
	x12	.512	.386	.447	.433	.430
	x13	.456	.358	.399	.388	.388
	x14	.443	.472	.401	.406	.426
	x15	.617	.488	.541	.527	.527
	x16	.612[a]	.411	.530	.506	.495
	x17	.411	.529[a]	.381	.397	.430
	x18	.530	.381	.461[a]	.443	.438
	x19	.506	.397	.443	.431[a]	.431
	x20	.495	.430	.438	.431	.438[a]
Residual[b]	x1	-.015	.021	-.068	-.024	-.031
	x2	-.016	-.036	.028	-.074	-.080
	x3	.027	-.001	.026	.042	-.069
	x4	-.036	.009	-.014	-.057	.096
	x5	-.007	.044	.032	-.042	-.047
	x6	-.016	-.014	-.028	-.079	.078
	x7	-.036	-.023	.004	.019	-.004
	x8	-.026	-.026	-.011	-.013	-.019
	x9	.007	.002	.011	-.016	-.021
	x10	.035	.028	.009	.087	-.030
	x11	.008	-.026	.030	-.039	-.030
	x12	.022	.026	-.020	-.054	.082
	x13	-.054	-.051	.103	.068	-.036
	x14	-.007	-.067	-.010	-.031	.077
	x15	.034	.035	-.021	.020	.033
	x16		.043	-.021	.035	-.004
	x17	.043		-.032	.064	-.034
	x18	-.021	-.032		.040	-.012
	x19	.035	.064	.040		.029
	x20	-.004	-.034	-.012	.029	

Extraction Method: Maximum Likelihood.

a. Reproduced communalities

b. Residuals are computed between observed and reproduced correlations. There are 32 (16.0%) nonredundant residuals with



Factor Analysis

Goodness-of-fit Test

Chi-Square	df	Sig.
287.501	151	.000

Pattern Matrix[a]

	Factor	
	1	2
x1	.565	.086
x2	.665	-.041
x3	.144	.489
x4	.408	.310
x5	-.163	.820
x6	.160	.406
x7	.209	.575
x8	.746	-.044
x9	.757	-.098
x10	.493	.017
x11	.777	-.033
x12	.622	.054
x13	.527	.089
x14	.299	.423
x15	.707	.130
x16	.830	-.072
x17	.120	.640
x18	.675	.006
x19	.585	.098
x20	.500	.210

Extraction Method: Maximum Likelihood.
Rotation Method: Oblimin with Kaiser Normalization.

a. Rotation converged in 6 iterations.

Structure Matrix

	Factor 1	Factor 2
x1	.624	.472
x2	.637	.413
x3	.478	.587
x4	.620	.589
x5	.398	.709
x6	.437	.515
x7	.602	.718
x8	.715	.465
x9	.689	.419
x10	.504	.354
x11	.754	.497
x12	.659	.479
x13	.588	.449
x14	.588	.627
x15	.796	.614
x16	.781	.496
x17	.557	.722
x18	.679	.467
x19	.653	.499
x20	.644	.552

Extraction Method: Maximum Likelihood.
Rotation Method: Oblimin with Kaiser Normalization.

Factor Correlation Matrix

Factor	1	2
1	1.000	.684
2	.684	1.000

Extraction Method: Maximum Likelihood.
Rotation Method: Oblimin with Kaiser Normalization.